

Project Overview

The goal of this project is to develop an AI-powered fantasy football system that uses historical and weekly NFL data to predict player performance and assist users with fantasy football decisions. The system will combine machine learning, deep learning, explainable AI, and retrieval-augmented generation (RAG).

Main Task 1 — NFL Data Collection and Feature Engineering

Estimated time to complete: 2-3 weeks

Develop an automated data pipeline for collecting and processing historical and weekly NFL player data for quarterbacks, running backs, wide receivers, and tight ends.

The dataset will include features such as:

- Weekly player statistics
- Fantasy points
- Targets and receptions
- Carries and rushing attempts
- Passing, rushing, and receiving yards
- Touchdowns
- Recent performance averages
- Player usage trends
- Opponent defensive performance
- Home/away status
- Team and game information

Historical data will be used to train the models, while new weekly NFL results will be incorporated to generate predictions for upcoming games.

Goal: Produce a structured player-by-week dataset that can automatically be updated throughout the NFL season.

Main Task 2 — Machine Learning Fantasy Point Prediction

Estimated time to complete: 2-3 weeks

Develop machine learning models that predict the fantasy points of NFL players for an upcoming week.

Several approaches will be evaluated, including:

- Statistical baseline models
- Linear Regression
- Random Forest
- XGBoost or similar gradient-boosting methods

Models will be evaluated using metrics such as:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2

Separate models may be developed for different positions because the factors affecting quarterback, running back, wide receiver, and tight end performance differ.

Goal: Determine which machine-learning approach provides the most accurate weekly fantasy football projections.

Main Task 3 — Deep Learning Sequential Prediction

Estimated time to complete: 3-4 weeks

Develop a deep-learning model that uses a player's recent sequence of games to predict future fantasy performance.

An LSTM or similar recurrent neural network will use several previous weeks of player performance as a sequence. Input features may include fantasy points, targets, carries, snap percentage, yards, touchdowns, and other usage statistics.

Example: Weeks 1-5 player performance → Deep Learning Model → Week 6 prediction

The deep-learning model will be compared against the traditional machine-learning models developed in Task 2.

Goal: Determine whether a model that learns temporal patterns in player performance can outperform traditional machine-learning approaches.

Main Task 4 — Explainable Start/Sit Recommendation System

Estimated time to complete: 3 weeks

Combine the best-performing prediction model with an explainable AI system that can assist with fantasy football start/sit decisions.

The system will compare players using:

- Predicted fantasy points
- Recent performance
- Matchup difficulty
- Player usage
- Prediction confidence
- Other important model features

Explainability techniques such as SHAP will be used to identify which features contributed most strongly to each prediction.

Example: Start Player A over Player B. Player A is projected for 18.4 fantasy points compared with 14.7 for Player B. Player A's recent target volume, favorable defensive matchup, and increased usage

were major contributors to the higher projection.

Goal: Transform the prediction models into an interpretable AI decision-support system rather than only producing numerical predictions.

Main Task 5 — RAG and LLM Fantasy Football Assistant

Estimated time to complete: 3 weeks

Develop a Retrieval-Augmented Generation system that supplements the numerical prediction model with current textual information such as injury reports, player news, depth-chart changes, and other relevant information.

Users could ask questions such as:

- “Should I start Player A or Player B?”
- “Why is Player A projected higher this week?”
- “What injury concerns should I know about?”

The LLM would combine information retrieved by RAG with predictions generated by the project's machine-learning models.

Main Task 6 — Boom/Bust and Risk Prediction

Estimated time to complete: 1 week

Expand the system beyond predicting a single expected fantasy-point value by estimating the probability of different player outcomes.

For example:

- Bust: Less than 10 points — 20%
- Normal: 10-20 points — 48%
- Boom: More than 20 points — 32%

This would allow the recommendation system to consider risk and upside in addition to expected fantasy points.

The system could potentially recommend different players depending on whether a fantasy manager needs a safe performance or a high-upside performance.

Additional / Stretch Tasks

Additional Task 1 — Transformer-Based Player Prediction

Develop a Transformer-based sequence model as an alternative to the LSTM.

The Transformer would analyze sequences of previous player performances and could potentially identify longer-term patterns in player usage and production.

Its performance would be compared against:

Baseline → Traditional ML → LSTM → Transformer

This would allow further investigation into whether increasingly complex deep-learning architectures provide meaningful improvements for fantasy football prediction.

Additional Task 2 — Waiver Wire Recommendation System

Develop a model for identifying players whose future fantasy value may be increasing.

The system could analyze:

- Increasing snap percentage
- Increasing targets
- Increasing carries
- Recent performance
- Injuries to teammates
- Changes in player roles
- Upcoming schedule difficulty

Players could then be ranked by predicted future fantasy value or breakout probability.

Goal: Extend the system from weekly start/sit decisions into proactive fantasy roster management.